PROJEK LEBUHRAYA PAN BORNEO SABAH FASA 1B
MEMBINA JALAN DARI SARANG KE TEMUNO (WP12)

Setting-out coordinate check (horizontal / level)

SETTING-OUT CHECK RECORD

REF. NO. **WP12-SOC-CST-SSU-20261019-1** | DATE **2026-10-19** | REVISION **0**

RESULT

FAIL

11 of 12 checked points PASS - 1 FAIL

12	11	1	30.1	11.4	+8.0
POINTS	PASS	FAIL	MAX RADIAL MM (P10)	RMS RADIAL MM	MAX DZ MM (P10)

CHECK DETAILS

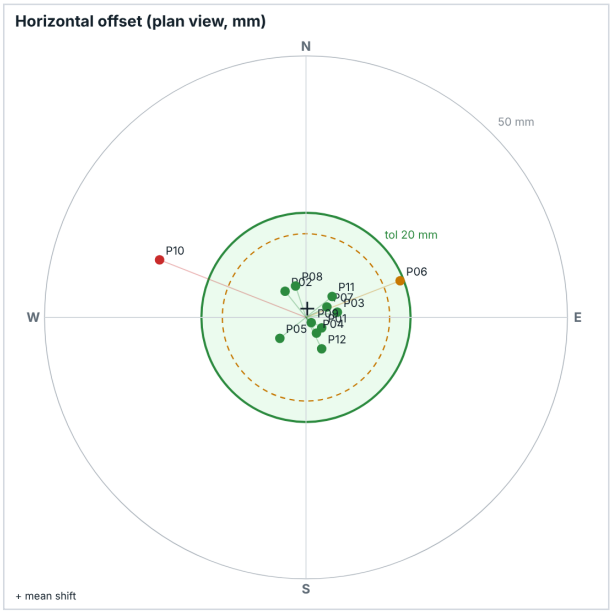
REF. NO.	WP12-SOC-CST-SSU-20261019-1	DATE	2026-10-19
RFI NO.	JKR-RFI-WP12-SUR-014	TEAM	CST
STRUCTURE	-	CHAINAGE / LOCATION	-
ELEMENT / DESCRIPTION	Pile cap PC-2 - bored pile positions	CHECK TYPE	Horizontal + Level
TOLERANCE	H 20 mm (radial) · RL +10 / -10 mm	COORDINATE SYSTEM	GDM2000 / RSO Borneo (BRSO)
TOLERANCE SOURCE	Sample ITP-07 rev A (fictional)	INSTRUMENT / S/N	Sample TS-100 S/N SN12345
STATION / BACKSIGHT	CP12 / CP11	CALIBRATION DATE	2026-08-01
BM / TBM USED	TBM7 RL 25.432		-

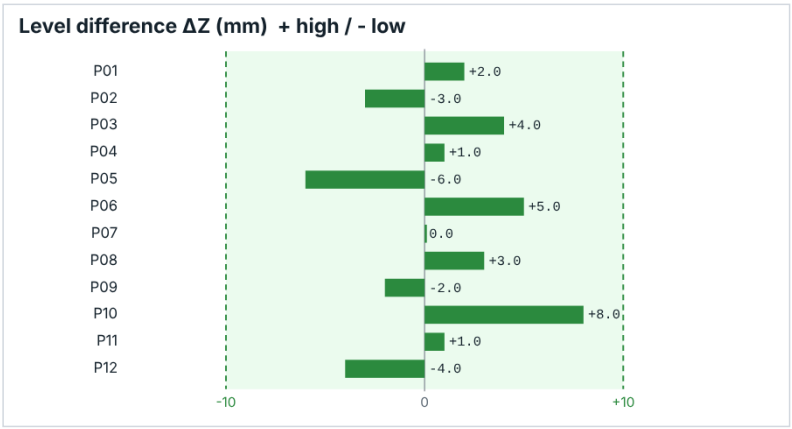
RESULTS

No	Point	Design			Measured			Offset (mm)				Result	Remarks
		E (X)	N (Y)	RL	E (X)	N (Y)	RL	dE	dN	Radial	dZ		
1	P01	612345.000	654321.000	24.500	612345.003	654320.998	24.502	+3.0	-2.0	3.6	+2.0	PASS	Within tolerance
2	P02	612347.400	654321.000	24.520	612347.396	654321.005	24.517	-4.0	+5.0	6.4	-3.0	PASS	Within tolerance
3	P03	612349.800	654321.000	24.540	612349.806	654321.001	24.544	+6.0	+1.0	6.1	+4.0	PASS	Within tolerance
4	P04	612352.200	654321.000	24.560	612352.202	654320.997	24.561	+2.0	-3.0	3.6	+1.0	PASS	Within tolerance
5	P05	612345.000	654323.400	24.580	612344.995	654323.396	24.574	-5.0	-4.0	6.4	-6.0	PASS	Within tolerance
6	P06	612347.400	654323.400	24.600	612347.418	654323.407	24.605	+18.0	+7.0	19.3	+5.0	PASS*	H within tolerance (97% of 20 mm)
7	P07	612349.800	654323.400	24.620	612349.804	654323.402	24.620	+4.0	+2.0	4.5	0.0	PASS	Within tolerance
8	P08	612352.200	654323.400	24.640	612352.198	654323.406	24.643	-2.0	+6.0	6.3	+3.0	PASS	Within tolerance
9	P09	612345.000	654325.800	24.660	612345.001	654325.799	24.658	+1.0	-1.0	1.4	-2.0	PASS	Within tolerance
10	P10	612347.400	654325.800	24.680	612347.372	654325.811	24.688	-28.0	+11.0	30.1	+8.0	FAIL	H exceeds tolerance by 10.1 mm - move 28 mm E, 11 mm S
11	P11	612349.800	654325.800	24.700	612349.805	654325.804	24.701	+5.0	+4.0	6.4	+1.0	PASS	Within tolerance
12	P12	612352.200	654325.800	24.720	612352.203	654325.794	24.716	+3.0	-6.0	6.7	-4.0	PASS	Within tolerance

dE = Measured E - Design E, dN = Measured N - Design N, Radial = sqrt(dE^2 + dN^2). dZ = Measured RL - Design RL (+ high / - low). PASS* = within tolerance but over 80% of it. Coordinates in metres, offsets in mm.

POINT ERROR DISTRIBUTION





SUMMARY

Points	**12**
Pass	**11 (1 marginal)**
Fail	**1**

HORIZONTAL (MM)

Max radial	**30.1 (P10)**
Mean radial	**8.4**
RMS radial	**11.4**
Mean shift dE/dN	**+0.3 / +1.7**

LEVEL (MM)

Max dZ	**+8.0 (P10)**
Mean dZ	**+0.8**
RMS dZ	**3.9**

SIGN-OFF

PREPARED BY

SSU

Surveyor (CST)

SIGNATURE (OPTIONAL) · 2026-10-19

CHECKED BY

SCH

Senior Surveyor

SIGNATURE (OPTIONAL) · 2026-10-19

Scan to open the app

Generated document — no signature required. Physical signatures may be added after printing if required.

Checking aid only: this report is generated from user-provided inputs and is not acceptance or certification. The user is responsible for independently verifying source data, point identity, units, coordinate and vertical datums, tolerances and results against approved project documents and field observations before use.

SITE PHOTOS (1-2 / 4)

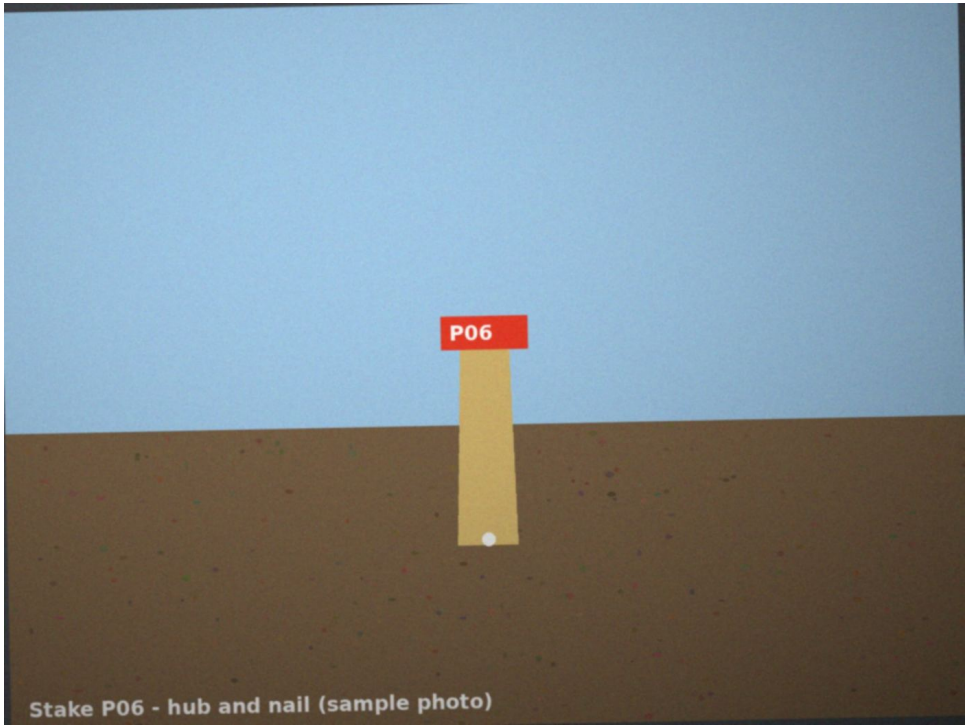


Photo 1
Stake P06 hub

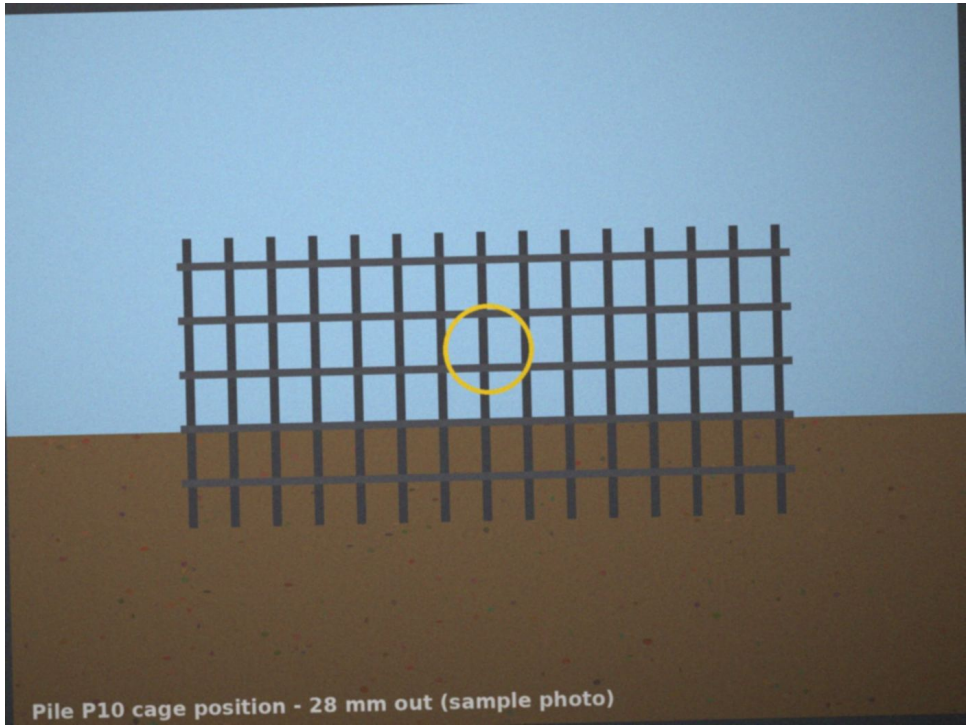


Photo 2
P10 cage 28 mm out

SITE PHOTOS (3-4 / 4)

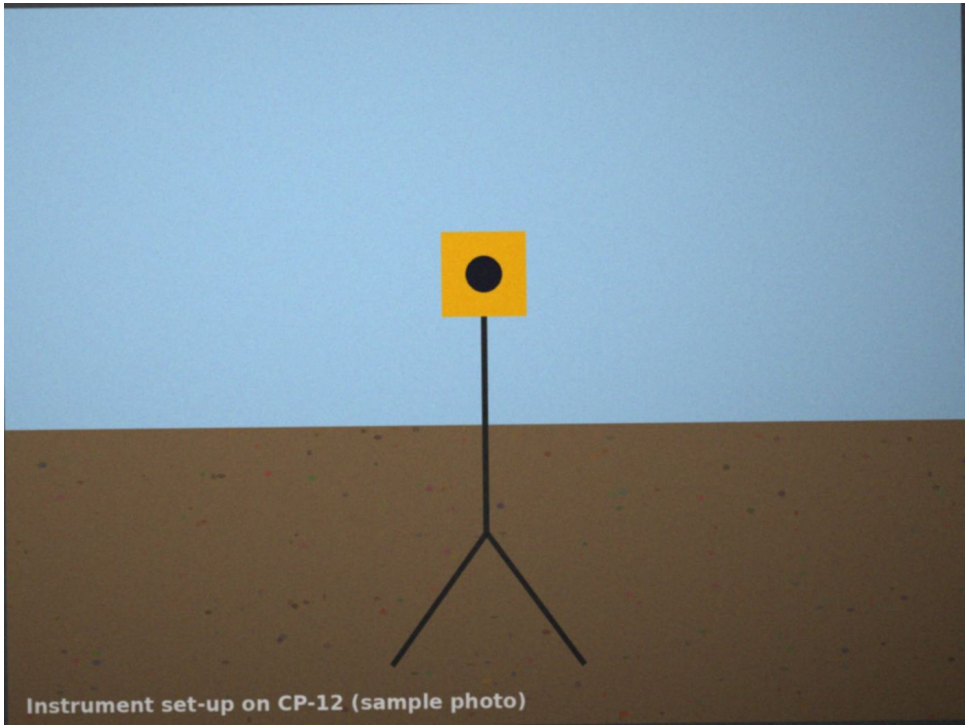


Photo 3
Instrument on CP12

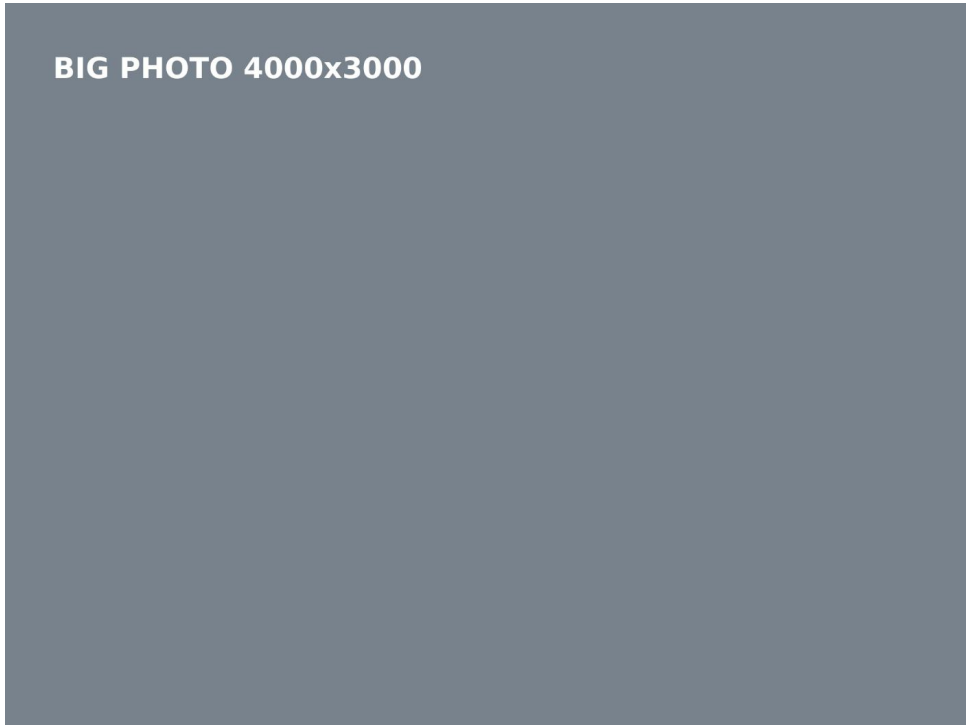


Photo 4
Oversize test